

Competing Risks and Multi-State Models

Concepts, methods and software

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Part V

Software



Contents:

1. Nonparametric estimation and plotting (Kaplan-Meier, Aalen-Johansen)
2. Creating data set with time-dependent censoring weights
3. PL-estimator and Fine and Gray model
4. Prediction

Outline

Nonparametric estimation: Aalen-Johansen

Single event type

Aalen-Johansen estimator

Nonparametric Estimation of Subdistribution

Time-varying values; counting process format

Representation of the weights

Creation of the data set with weights

The product-limit estimator

Log-rank tests

Fine & Gray model

Prediction

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Right Censored Data

Representation Two columns: **time** since origin and **status** variable. `status=1` if event observed and `status=0` if event right censored

```
id time status
1  7.7      1
2  4.3      1
3  5.6      0
...
```

In R via `Surv(time=time, event=status)`

Right Censored Data

Representation Two columns: **time** since origin and **status** variable. `status=1` if event observed and `status=0` if event right censored

```
id time status
1  7.7      1
2  4.3      1
3  5.6      0
...
```

In R via `Surv(time=time, event=status)`

Kaplan-Meier Main function: `survfit.formula`

```
survfit(Surv(time,status)~1, data=...)
```

Four example individuals

aidssi data set (available in `mstate` package and in Stata)

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

Four example individuals

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14	5.054	0	event-free	WW
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Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time, status!=0)~1,
                      data=aidssi)
```

Note: `event` argument can be a logical expression
(`status!=0`)

Four example individuals

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patnr	time	status	cause	ccr5
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Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time, status!=0)~1,
                      data=aidssi)
```

Note: event argument can be a logical expression
(status!=0)

Kaplan-Meier per value of CCR5:

```
KM.ccr5 <- survfit(Surv(time, status!=0)~ccr5,
                   data=aidssi)
```

Numerical summary

Basic summary:

```
KM.overall
```

	n	events	median	0.95LCL	0.95UCL
[1,]	329	222	7.3	6.44	8.36

More detailed summary:

```
summary(KM.overall)
```

time	n.risk	n.event	survival	std.err	lower	95% CI	upper	95% CI
0.112	329	1	0.997	0.00303		0.991		1.000
0.137	328	1	0.994	0.00429		0.986		1.000
0.474	325	1	0.991	0.00525		0.981		1.000
.	.	.	.	.		.		.
12.936	41	1	0.217	0.02604		0.171		0.274
13.361	22	1	0.207	0.02665		0.161		0.266
13.936	1	1	0.000	NaN		NA		NA

Survival at 12 years obtained via

```
summary(KM.overall, time=12)
```

In R

Uses the function `survfit.formula`, which calls `survfitKM`

```
survfitKM
function (x, y, weights = rep(1, length(x)),
  stype = 1, ctype = 1,
  se.fit = TRUE, conf.int = 0.95,
  conf.type = c("log", "log-log", "plain", "none",
                "logit", "arcsin"),
  conf.lower = c("usual", "peto", "modified"),
  start.time, id, cluster, robust,
  influence = FALSE, type, entry = FALSE, time0 = FALSE)
```

Plotting Kaplan-Meier survival curves

`plot.survfit` for first plot:

```
function (x, conf.int,
  mark.time = FALSE, pch = 3, col = 1, lty = 1, lwd = 1, cex = 1,
  log = FALSE, xscale = 1, yscale = 1, xlim, ylim,
  xmax, fun, xlab = "", ylab = "", xaxs = "r",
  conf.times, conf.cap = 0.005, conf.offset = 0.012,
  conf.type = c("log", "log-log", "plain", "logit", "arcsin", "none"),
  mark.col, noplot = "(s0)", cumhaz = FALSE, firstx, ymin,
  cumprob = FALSE, ...)
```

`fun="event"` plots cumulative incidence (i.e. upwards from 0):

```
plot(KM.overall, fun="event")
```

`lines.survfit` adds curves

ggsurvfit package

- Extension of `ggplot2`.
See <https://www.danielsjoberg.com/ggsurvfit>
- Allows for number-at-risk table, automatic legend
- Basic survival function via

```
ggsurvfit(survfit2(Surv(time, status!=0)~ccr5,
                  data=aidssi))
```

Use `survfit2` for nicer labeling of groups

- `type="risk"` plots cumulative incidence
- CIs not plotted by default; use
`+ add_confidence_interval()`

survminer package

- Extension of `ggplot2`.

See [https:](https://rpkgs.datanovia.com/survminer/index.html)

[//rpkgs.datanovia.com/survminer/index.html](https://rpkgs.datanovia.com/survminer/index.html)

- Allows for number-at-risk table, automatic legend
- Basic survival function via

```
ggsurvplot(survfit(Surv(time, status!=0)~ccr5,
                    data=aidssi))
```

- `fun="event"` plots cumulative incidence

Four example individuals

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Kaplan-Meier both event types combined:

```
KM.overall <- survfit(Surv(time, status!=0)~1,
                      data=aidssi)
```

Note: event argument can be a logical expression
(status!=0)

Kaplan-Meier per value of CCR5:

```
KM.ccr5 <- survfit(Surv(time, status!=0)~ccr5,
                   data=aidssi)
```

Results Cox model for combined end point

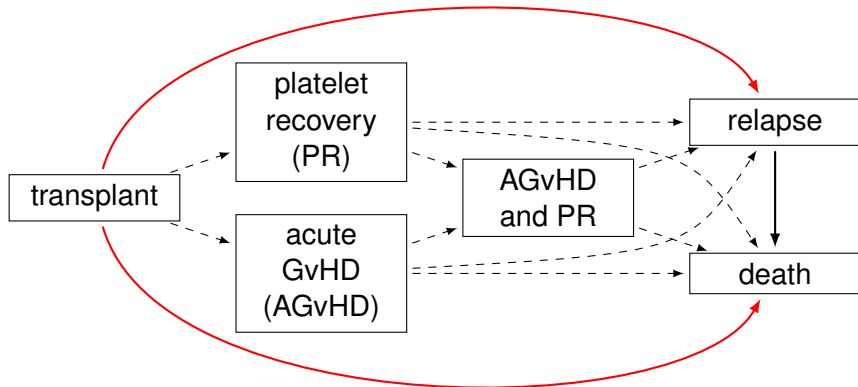
```
coxph(formula = Surv(time, status != 0) ~ ccr5,
      data = aidssi)
```

Result:

	coef	exp(coef)	se(coef)	z	p
ccr5WM	-0.701	0.496	0.186	-3.77	0.00016

Likelihood ratio test=16.5 on 1 df, p=4.95e-05 n= 324,
 number of events= 220
 (5 observations deleted due to missingness)

Computer practical



1. **A first look at the data** Have a look at the data explanation:
`help(ebmt1)`
Inspect the data.
Summarize all the variables. What do you observe?
2. **Estimation of overall survival** Compute the Kaplan-Meier estimator for survival. Give a numerical summary and plot the estimate using the `survival`, `ggsurvfit` and `survminer` packages.
3. **A Cox model for overall survival** Compute and plot the Kaplan-Meier estimator for survival by value of **score**, using the same three packages as in 2. Fit a Cox model for the effect of the EBMT score on survival.

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Software to compute Aalen-Johansen estimator

- Stata: `stcompet` command
- SAS: macros `%CUMINCID`
or `%CIF`
- R, some options:
 - standard `survival` package
 - `cmprsk` or `prodlim` package
 - any package for multi-state models, e.g. `etm`, `mstate`, `msSurv`

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- Stata: `stcompet` command
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- R, some options:
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 - any package for multi-state models, e.g. `etm`, `mstate`, `msSurv`

Aalen-Johansen estimator

Event column needs to be a *factor*:

```
aidssi$cause <- relevel(aidssi$cause, ref="event-free")
csiSurv <- survfit(Surv(time, event=cause)~1,
                  data=aidssi)
```

Summary of estimate:

```
summ <- summary(csiSurv, times=seq(2,8,by=2))
```

summ

time	n.risk	n.event	Pr((s0))	Pr(AIDS)	Pr(SI)
2	300	15	0.953	0.00632	0.0404
4	240	52	0.786	0.10322	0.1112
6	170	54	0.603	0.17070	0.2259
8	122	41	0.454	0.25430	0.2916

Confidence intervals: `summ$lower` and `summ$upper`

Plotting

- **survival: use same plot function**
 - specific event types selected via `csiSurv["AIDS"]` etc.
 - default curve shown from 0 upwards `fun="event"`
- **ggsurvfit: use `ggcuminc` function**
 - specific event types selected via `outcome` argument
 - alternate and stacked display format not implemented

```
ggcuminc(csiSurv, outcome=c("AIDS", "SI"))
```

- **survminer: use `ggcompetingrisks` function**
 - less flexible than `ggcuminc` in `ggsurvfit`

5. **Estimation of cause-specific cumulative incidence** Compute the Aalen-Johansen estimator for relapse and relapse-free mortality. What is the probability, with 95% confidence intervals (on the default log scale), to have a relapse within one year and within five years.
Compare the estimates at 1 and 5 years with the relapse-free survival distribution, i.e. the distribution of time to remain free of both event types.
6. **Some plots** (a) Plot the estimated cause-specific cumulative incidence for each end point using the overlaid display format. Plot the 95% confidence intervals as well.
(b) Plot the estimated cause-specific cumulative incidence using the stacked format (without the confidence intervals).
(c) Plot the cause-specific cumulative incidence estimates for each end point using the alternate display format.
7. **Regression on the cause-specific hazards** Fit a proportional hazards model for relapse and survival as competing risks.

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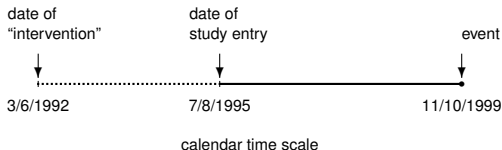
Log-rank tests

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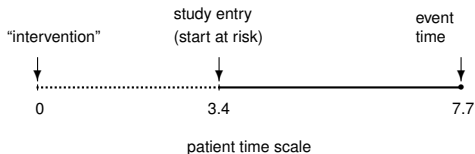
Prediction

Left truncated data: structure

- In calendar time scale



- In patient time scale



Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

- **Data:** extra column, describing entry time in risk set

id	entry time	event time	status
1	0.0	4.3	1
2	0.0	5.6	0
3	3.4	7.7	1

- In R: `Surv(entry.time, event.time, status)`

Left truncated data: analysis and data representation

Individuals only contribute while they are in follow-up

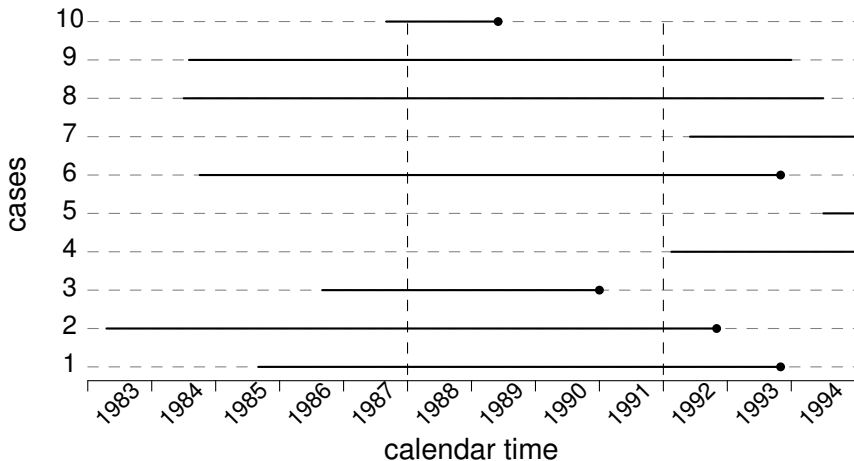
- **Data:** extra column, describing entry time in risk set

id	entry time	event time	status
1	0.0	4.3	1
2	0.0	5.6	0
3	3.4	7.7	1

- In R: `Surv(entry.time, event.time, status)`
- Note: log-rank test function `survdif` does not work with left-truncated data; use equivalence with score test from Cox model

```
> coxph(  
  Surv(entry.time, event.time, status) ~ ccr5,  
  ...) $score
```

Example time-varying covariable



Data format

- start time, stop time and event status at stop time, plus covariable value in interval.
Represented by three rows:

id	Tstart	Tstop	status	cal.period
1	0	2.3	0	< 1988
1	2.3	6.3	0	1988-1991
1	6.3	8.2	1	> 1991

Enters in risk set “1988-1991” (“> 1991”) after 2.3 (6.3) years

Data format

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id	Tstart	Tstop	status	cal.period
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1	2.3	6.3	0	1988-1991
1	6.3	8.2	1	> 1991

Enters in risk set “1988-1991” (“> 1991”) after 2.3 (6.3) years

- R: `Surv(Tstart, Tstop, status) ~ cal.period`

Data format

- start time, stop time and event status at stop time, plus covariable value in interval.
Represented by three rows:

id	Tstart	Tstop	status	cal.period
1	0	2.3	0	< 1988
1	2.3	6.3	0	1988-1991
1	6.3	8.2	1	> 1991

Enters in risk set “1988-1991” (“> 1991”) after 2.3 (6.3) years

- $R: \text{Surv}(T_{\text{start}}, T_{\text{stop}}, \text{status}) \sim \text{cal.period}$
- Similar for discontinuous risk intervals and for time updated weights

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Example data

Three individuals from data set

id	entry.time	event.time	event.type
1	0.25460	0.63644	0
2	0.00000	0.64358	1
3	0.08005	0.25615	2

General product-limit estimator of F_k

$$\widehat{F}_k^{\text{PL}}(t) = \prod_{i: t_{(i)} \leq t} \left\{ 1 - \widehat{h}_k(t_{(i)}) \right\}$$

$$\widehat{h}_k(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1
- competing event at $t_{(j)}$ before $t_{(i)}$: weight

$$\frac{\widehat{\Gamma}(t_{(i)}-)}{\widehat{\Gamma}(t_{(j)}-)} \times \frac{\widehat{\Phi}(t_{(i)}-)}{\widehat{\Phi}(t_{(j)}-)}$$

Example data

Three individuals from data set

id	entry.time	event.time	event.type
1	0.25460	0.63644	0
2	0.00000	0.64358	1
3	0.08005	0.25615	2



id	Tstart	Tstop	status	weight.cens	weight.trunc	count	failcode
1	0.25460	0.63644	0	1.00000	1.00000	1	1
2	0.00000	0.64358	1	1.00000	1.00000	1	1
3	0.08005	0.25615	2	1.00000	1.00000	1	1
3	0.25615	0.31778	2	1.00000	1.02941	2	1
3	0.31778	0.37693	2	1.00000	1.02941	3	1
3	0.37693	0.38928	2	1.00000	1.02941	4	1
3	0.38928	0.46029	2	1.00000	1.02941	5	1
3	0.46029	0.50979	2	1.00000	1.02941	6	1
3	0.50979	0.64358	2	0.67849	1.07230	7	1
3	0.64358	0.64724	2	0.67849	1.07230	8	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Events of type 1 observed at 0.31778 0.37693 0.38928 0.46029 0.50979 0.64358 0.64724

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Software

Stata `stcrprep` command

SAS `%PSHREG` macro

R `finegray` function in `survival` package
`crprep` function in `mstate` package

R: create data set with weights

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

```
finegray(Surv(time,cause)~., data=aidssi, etype="AIDS")
```

Note: event-free first level in factor specification

```
crprep(Tstop="time", status="status", data=aidssi,  
       trans=1, cens=0, keep="ccr5")  
aidssi.w <- crprep(Tstop="time", status="cause",  
                  data=aidssi, trans="AIDS", cens="event-free",  
                  keep="ccr5")
```

Both event types at once via `trans=c("AIDS", "SI")`

Resulting data set (crprep)

patnr	time	status	cause	ccr5
14	5.054	0	event-free	WW
3	2.234	1	AIDS	WW
15	10.196	1	AIDS	WM
8	8.605	2	SI	WW

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
3	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
17	8	0.000	8.605	SI	1.000	WW	1	AIDS
18	8	8.605	8.638	SI	0.991	WW	2	AIDS
19	8	8.638	8.755	SI	0.982	WW	3	AIDS
.	.	.	.	.	.	.	.	.
78	14	0.000	5.054	event-free	1.000	WW	1	AIDS
79	15	0.000	10.196	AIDS	1.000	WM	1	AIDS

finegray:

- different column names **patnr**, **fgstart**, **fgstop**, **fgwt**
- **status** converted to numeric; extra **fgstatus** 0-1 column
- no **failcode** column, **count** column not default

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PL-form: Kaplan-Meier with probability weights

Stata Specify `pweights` option in `stset` command;
use standard `sts` command

SAS PROC LIFEREG

R `weights` argument in `survfit` function
(`survival` package)

```
survfit(Surv(Tstart, Tstop, status=="AIDS")~1,
        data=aidssi.w, weights=weight.cens)
```

PL-form: Kaplan-Meier with probability weights

Stata Specify `pweights` option in `stset` command;
use standard `sts` command

SAS PROC LIFEREG

R `weights` argument in `survfit` function
(`survival` package)

```
survfit(Surv(Tstart, Tstop, status=="AIDS")~1,
        data=aidssi.w, weights=weight.cens)
```

- Additional left truncation:

```
weights=weight.cens*weight.trunc
```

Both event types at once: resulting data set

	id	Tstart	Tstop	status	weight.cens	ccr5	count	failcode
	3	0.000	2.234	AIDS	1.000	WW	1	AIDS
	17	0.000	8.605	SI	1.000	WW	1	AIDS
	18	8.605	8.638	SI	0.991	WW	2	AIDS
	19	8.638	8.755	SI	0.982	WW	3	AIDS
	.	.	.	.	.	.	.	.
	78	0.000	5.054	event-free	1.000	WW	1	AIDS
	79	0.000	10.196	AIDS	1.000	WM	1	AIDS
	.	.	.	.	.	.	.	.
	.	.	.	.	.	.	.	.
	2768	0.000	2.234	AIDS	1.000	WW	1	SI
	2769	2.234	2.322	AIDS	0.997	WW	2	SI
	2770	2.322	2.982	AIDS	0.993	WW	3	SI
	.	.	.	.	.	.	.	.
	2870	0.000	8.605	SI	1.000	WW	1	SI
	2913	0.000	5.054	event-free	1.000	WW	1	SI
	2914	0.000	10.196	AIDS	1.000	WM	1	SI
	2915	10.196	10.448	AIDS	0.974	WM	2	SI
	2916	10.448	10.467	AIDS	0.961	WM	3	SI
	.	.	.	.	.	.	.	.

Both event types at once: estimation

```
csiPL <- survfit(  
  Surv(Tstart,Tstop,status==failcode)~failcode,  
  data=aidssi.w, weights=weight.cens)  
csiPL$strata  
## failcode=AIDS    failcode=SI  
##              310              310  
summary(csiPL["failcode=SI"],times=seq(2,10,by=2))
```

and obtain

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
2	302	13	0.959615	0.0110	0.938344	0.981369
4	272	22	0.888783	0.0177	0.854687	0.924239
6	218	34	0.774112	0.0240	0.728466	0.822619
8	190	18	0.708440	0.0265	0.658364	0.762324
10	161	11	0.665410	0.0279	0.612936	0.722377

or use `csiPL[2]`

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Gray's log-rank test

- Implemented directly, using original (unweighted) data set

SAS: %CIF macro

R: `cuminc` function in `cmprsk` package

- Use score test from proportional subdistribution hazards model
 - Compute censoring weights per subgroup (e.g. via `strata` argument in `crprep` function)

```
aidssi.wCCR <- crprep(Tstop="time",
  status="cause", data=aidssi,
  trans=c("AIDS", "SI"), cens="event-free",
  id="patnr", strata="ccr5")
```

- Variance of the test statistic is different from the one suggested by Gray. Affects p-value

R example code

```
test.AIDS <- coxph(Surv(Tstart,Tstop,status=="AIDS")~ccr5,
                  data=aidssi.wCCR, weights=weight.cens,
                  subset=failcode=="AIDS")
test.SI <- coxph(Surv(Tstart,Tstop,status=="SI")~ccr5,
                data=aidssi.wCCR, weights=weight.cens,
                subset=failcode=="SI")
```

Score test statistic and p-value

```
## AIDS
c(test.AIDS$score, 1-pchisq(test.AIDS$score,1))
[1] 12.428139562 0.000422913
## SI
c(test.SI$score, 1-pchisq(test.SI$score,1))
[1] 0.007207246 0.932344471
```

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Fine & Gray model

Prediction

Proportional subdistribution hazards (Fine & Gray)

- Implemented directly (using robust estimator of standard error; no left truncation)

Stata `stcrreg` command

SAS `PROC PHREG` procedure (SAS/STAT ≥ 13.1)

R `crr` function in `cmprsk` package;
FGR function from `riskRegression`
package provides a formula interface

Proportional subdistribution hazards (Fine & Gray)

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 - SAS `PROC PHREG` procedure (SAS/STAT ≥ 13.1)
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- Translate to cumulative incidence: use standard code for prediction based on results of Cox model with weights

8. **Estimate weights and compute PL-estimates.** Compute the data set with weights for relapse. Include **type** and **type** and the covariables **score** and **age**. Inspect the data. Compute and plot the PL-form for relapse and add AJ-form. Compare estimates and 95% CIs.
9. **Estimate per **score** value, based on PL-estimate** Compute the weights and the PL-estimates. Plot the results, using the overlaid, stacked and alternate formats. Compute weights by value of **score**? Estimate and plot the time-to-censoring distribution by **score**.
10. **Log-rank tests** Run and compare log-rank tests
11. **Impact of EBMT score on subdistribution hazards**
Compare the parameter estimates with the ones from the proportional cause-specific hazards model. Can you explain the difference?

Outline

Nonparametric estimation: Aalen-Johansen

- Single event type

- Aalen-Johansen estimator

Nonparametric Estimation of Subdistribution

- Time-varying values; counting process format

- Representation of the weights

- Creation of the data set with weights

- The product-limit estimator

- Log-rank tests

Fine & Gray model

Prediction

From cause-specific hazard regression to cumulative incidence

Stata `stpm2` for parametric hazard model; `stpm2cif`
translates to cumulative incidence

SAS Macros `CumInc.sas` and `CumIncV.sas`

R recent version of `survival` package
`mstate` package

From Fine & Gray to cumulative incidence

```
fit.FG <- coxph(Surv(Tstart,Tstop,status==failcode) ~  
               ccr.comb + strata(failcode), data=aidssi.w,  
               robust=FALSE, weights=weight.cens)
```

Predict for four possible combinations

```
indivs <- data.frame(  
  ccr.comb=factor(c("WW", "WW", "WM.AIDS", "WM.SI")),  
  failcode=c("AIDS", "SI", "AIDS", "SI"))  
pred.sdh <- survfit(fit.FG, newdata=indivs)
```

Results per combination, e.g. for SI and WW:

```
pred.sdh[2]
```